

HW4

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Question 1

Lemma 0.1 (Relevance Lemma for FOL). *Let ϕ be a FOL formula, let $\mathfrak{A}$ be a structure and let α_1, α_2 be variable assignments such that for all $x \in FVar(\phi)$,*

$$\alpha_1(x) = \alpha_2(x).$$

Then

$$\mathfrak{A}, \alpha_1 \models \phi \iff \mathfrak{A}, \alpha_2 \models \phi.$$

Question. Prove Lemma 1.

Proof. By symmetry, we only need to prove if for all $x \in FVar(\phi), \alpha_1(x) = \alpha_2(x)$ then

$$\mathfrak{A}, \alpha_1 \models \phi \implies \mathfrak{A}, \alpha_2 \models \phi.$$

Lemma 0.2. *If*

$$\forall x \in Var(t), \alpha_1(x) = \alpha_2(x)$$

then $Val_{\mathfrak{A}, \alpha_1}(t) = Val_{\mathfrak{A}, \alpha_2}(t)$

Base case:

1. $t \equiv c$: $I(c) = I(c) \implies Val_{\mathfrak{A}, \alpha_1}(c) = Val_{\mathfrak{A}, \alpha_2}(c)$
2. $t \equiv x$: By definition, $Val_{\mathfrak{A}, \alpha_1}(x) = \alpha_1(x) = \alpha_2(x) = Val_{\mathfrak{A}, \alpha_2}(x)$

Inductive step

Let

$$t \equiv f(t_1, \dots, t_n).$$

$$Val_{\mathfrak{A}, \alpha_1}(t) = I_{\mathfrak{A}}(f)(Val_{\mathfrak{A}, \alpha_1}(t_1), \dots, Val_{\mathfrak{A}, \alpha_1}(t_n)).$$

By IH, for all i

$$\begin{aligned} Val_{\mathfrak{A}, \alpha_1}(t_i) &= Val_{\mathfrak{A}, \alpha_2}(t_i) \\ (Val_{\mathfrak{A}, \alpha_1}(t_1), \dots, Val_{\mathfrak{A}, \alpha_1}(t_n)) &= (Val_{\mathfrak{A}, \alpha_2}(t_1), \dots, Val_{\mathfrak{A}, \alpha_2}(t_n)). \end{aligned}$$

$$\begin{aligned} Val_{\mathfrak{A}, \alpha_1}(t) &= I_{\mathfrak{A}}(f)(Val_{\mathfrak{A}, \alpha_1}(t_1), \dots) \\ &= I_{\mathfrak{A}}(f)(Val_{\mathfrak{A}, \alpha_2}(t_1), \dots) \\ &= Val_{\mathfrak{A}, \alpha_2}(t). \end{aligned}$$

Now we can use the lemma to prove the previous lemma:

Base cases

1. $\phi \equiv t_1 = t_2$. Suppose

$$\mathfrak{A}, \alpha_1 \models t_1 = t_2.$$

Then we have

$$Val_{\mathfrak{A}, \alpha_1}(t_1) = Val_{\mathfrak{A}, \alpha_1}(t_2).$$

$$FVar(t_1 = t_2) = Var(t_1) \cup Var(t_2),$$

$$Val_{\mathfrak{A}, \alpha_1}(t_1) = Val_{\mathfrak{A}, \alpha_2}(t_1) \quad \text{and} \quad Val_{\mathfrak{A}, \alpha_1}(t_2) = Val_{\mathfrak{A}, \alpha_2}(t_2).$$

$$Val_{\mathfrak{A}, \alpha_2}(t_1) = Val_{\mathfrak{A}, \alpha_2}(t_2) \implies \mathfrak{A}, \alpha_2 \models t_1 = t_2.$$

2. $\phi \equiv R(t_1, \dots, t_k)$.

Suppose

$$\mathfrak{A}, \alpha_1 \models R(t_1, \dots, t_k).$$

Then

$$(Val_{\mathfrak{A}, \alpha_1}(t_1), \dots, Val_{\mathfrak{A}, \alpha_1}(t_k)) \in I(R).$$

$$FVar(R(t_1, \dots, t_k)) = Var(t_1) \cup \dots \cup Var(t_k),$$

by the term lemma,

$$\forall i, Val_{\mathfrak{A}, \alpha_1}(t_i) = Val_{\mathfrak{A}, \alpha_2}(t_i).$$

$$(Val_{\mathfrak{A}, \alpha_1}(t_1), \dots, Val_{\mathfrak{A}, \alpha_1}(t_k)) = (Val_{\mathfrak{A}, \alpha_2}(t_1), \dots, Val_{\mathfrak{A}, \alpha_2}(t_k)) \in I(R)$$

$$\mathfrak{A}, \alpha_2 \models R(t_1, \dots, t_k).$$

Inductive cases

1. $\phi \equiv \neg\psi$.

Suppose

$$\mathfrak{A}, \alpha_1 \models \neg\psi.$$

Then

$$\mathfrak{A}, \alpha_1 \not\models \psi.$$

$$FVar(\neg\psi) = FVar(\psi),$$

$$\mathfrak{A}, \alpha_2 \models \psi \implies \mathfrak{A}, \alpha_1 \models \psi.$$

$$\mathfrak{A}, \alpha_2 \not\models \psi,$$

$$\mathfrak{A}, \alpha_2 \models \neg\psi.$$

2. $\phi \equiv \psi_1 \vee \psi_2$.

$$\mathfrak{A}, \alpha_1 \models \psi_1 \vee \psi_2.$$

then either

$$\mathfrak{A}, \alpha_1 \models \psi_1 \quad \text{or} \quad \mathfrak{A}, \alpha_1 \models \psi_2.$$

Also,

$$FVar(\psi_1 \vee \psi_2) = FVar(\psi_1) \cup FVar(\psi_2).$$

by IH

$$\mathfrak{A}, \alpha_2 \models \psi_1 \quad \text{or} \quad \mathfrak{A}, \alpha_2 \models \psi_2.$$

Therefore,

$$\mathfrak{A}, \alpha_2 \models \psi_1 \vee \psi_2.$$

3. $\phi \equiv \exists x \psi$.

Suppose

$$\mathfrak{A}, \alpha_1 \models \exists x \psi.$$

Then there exists some $u \in U$ such that

$$\mathfrak{A}, \alpha_1[x \mapsto u] \models \psi.$$

w can show that $\alpha_1[x \mapsto u]$ and $\alpha_2[x \mapsto u]$ agree on all variables in $FVar(\psi)$.

fix a $y \in FVar(\psi)$,

- If $y = x$, then

$$\alpha_1[x \mapsto u](y) = u = \alpha_2[x \mapsto u](y).$$

- If $y \neq x$, then $y \in FVar(\psi) \setminus \{x\} = FVar(\exists x, \psi)$

$$\alpha_1[x \mapsto u](y) = \alpha_1(y) = \alpha_2(y) = \alpha_2[x \mapsto u](y).$$

Thus, $\forall y \in FVar(\psi)$,

$$\alpha_1[x \mapsto u](y) = \alpha_2[x \mapsto u](y).$$

By IH

$$\mathfrak{A}, \alpha_2[x \mapsto u] \models \psi.$$

Hence

$$\mathfrak{A}, \alpha_2 \models \exists x \psi.$$

4. $\phi \equiv \forall x \psi$.

Suppose

$$\mathfrak{A}, \alpha_1 \models \forall x \psi.$$

then for every $u \in U$,

$$\mathfrak{A}, \alpha_1[x \mapsto u] \models \psi.$$

Fix a $u \in U$. We can show that $\alpha_1[x \mapsto u]$ and $\alpha_2[x \mapsto u]$ agree on all variables in $FVar(\psi)$.

fix a $y \in FVar(\psi)$.

- If $y = x$, then

$$\alpha_1[x \mapsto u](y) = u = \alpha_2[x \mapsto u](y).$$

- If $y \neq x$, then $y \in FVar(\psi) \setminus \{x\} = FVar(\exists x, \psi)$

$$\alpha_1[x \mapsto u](y) = \alpha_1(y) = \alpha_2(y) = \alpha_2[x \mapsto u](y).$$

by IH,

$$\mathfrak{A}, \alpha_2[x \mapsto u] \models \psi.$$

Since $u \in U$ is arbitrary,

$$\mathfrak{A}, \alpha_2 \models \forall x \psi.$$

□

Question 2

Definition 0.3 (Ground Terms, Atoms and Formulae). *A term over a signature Σ is said to be a ground term if it does not contain any occurrence of a variable. The set of all ground terms over Σ is denoted by*

$$\text{Terms}_\Sigma.$$

A formula is said to be ground if it contains no variables (and therefore no quantifiers).

A formula is said to be a ground atom if it is both ground and atomic.

We denote by

$$\text{Atoms}_\Sigma$$

the set of all ground atoms, and by

$$\text{FORM}_\Sigma$$

the set of all ground formulae over Σ .

Definition 0.4 (Ground Instantiation). *Let*

$$\phi \equiv \forall x_1 \forall x_2 \cdots \forall x_n \psi$$

be a universally quantified FOL sentence in prenex normal form.

A ground instantiation of ϕ is obtained by replacing each occurrence of x_i in ψ by a ground term t_i .

Formally, there exists a mapping

$$T : V \rightarrow \text{Terms}_\Sigma$$

such that

$$\varphi \equiv \psi[x_1 \rightarrow T(x_1)] \cdots [x_n \rightarrow T(x_n)].$$

The set of all ground instantiations of ϕ is denoted

$$\text{Ground}(\phi).$$

For a set Γ of universally quantified sentences,

$$\text{Ground}(\Gamma) = \bigcup_{\phi \in \Gamma} \text{Ground}(\phi).$$

Definition 0.5 (Propositional Satisfiability). A ground atomic valuation over Σ is a mapping

$$\tau : \text{Atoms}_\Sigma \rightarrow \{\text{true}, \text{false}\}.$$

This valuation is extended to ground formulae using the relation $\models_{Gr}$ defined by:

$$\tau \models_{Gr} \psi \quad \text{iff} \quad \tau(\psi) = \text{true} \quad \text{if } \psi \in \text{Atoms}_\Sigma$$

$$\tau \models_{Gr} \neg\phi \quad \text{iff} \quad \tau \not\models_{Gr} \phi$$

$$\tau \models_{Gr} (\phi_1 \vee \phi_2) \quad \text{iff} \quad \tau \models_{Gr} \phi_1 \text{ or } \tau \models_{Gr} \phi_2.$$

A set of ground FOL formulae Γ is propositionally satisfiable if there exists a ground atomic valuation τ such that

$$\tau \models_{Gr} \phi \quad \text{for every } \phi \in \Gamma.$$

Lemma 0.6. If Γ is satisfiable then $\text{Ground}(\Gamma)$ is propositionally satisfiable.

Question. Prove Lemma 2.

Proof. Γ is satisfiable, there exists a structure $\mathfrak{A}$ such that

$$\mathfrak{A} \models \Gamma.$$

Define ground atom valuation

$$\tau_{\mathfrak{A}} : \text{Atoms}_\Sigma \rightarrow \{\text{true}, \text{false}\}$$

$$\tau_{\mathfrak{A}}(a) = \text{true} \iff \mathfrak{A} \models a$$

for every ground atom a over Σ . Now we show that for every ground formula ϕ ,

$$\mathfrak{A} \models \phi \implies \tau_{\mathfrak{A}} \models_{Gr} \phi.$$

Induction on the structure of ϕ

Base case: χ is a ground atom

trivial

Inductive cases

1. $\psi \equiv \neg\phi$. Suppose

$$\mathfrak{A} \models \neg\phi \implies \mathfrak{A} \not\models \phi$$

By IH,

$$\mathfrak{A} \models \phi \implies \tau_{\mathfrak{A}} \models_{Gr} \phi$$

$$\tau_{\mathfrak{A}} \not\models_{Gr} \phi$$

$$\tau_{\mathfrak{A}} \models_{Gr} \neg\phi$$

2. $\chi \equiv \phi_1 \vee \phi_2$.

Suppose

$$\mathfrak{A} \models \phi_1 \vee \phi_2$$

Then

$$\mathfrak{A} \models \phi_1 \quad \text{or} \quad \mathfrak{A} \models \phi_2$$

by IH

$$\tau_{\mathfrak{A}} \models_{Gr} \phi_1 \quad \text{or} \quad \tau_{\mathfrak{A}} \models_{Gr} \phi_2$$

$$\tau_{\mathfrak{A}} \models_{Gr} \phi_1 \vee \phi_2$$

Fix any $\chi \in \text{Ground}(\Gamma)$. Then there exists

$$\varphi = \forall x_1 \cdots \forall x_n \psi \in \Gamma$$

such that

$$\chi = \psi[x_1 \mapsto t_1] \cdots [x_n \mapsto t_n]$$

for ground terms $t_1, \dots, t_n$

$$\mathfrak{A} \models \Gamma \implies \mathfrak{A} \models \varphi$$

For $1 \leq i \leq n$

$$u_i = \text{Val}_{\mathfrak{A}}(t_i)$$

$$\mathfrak{A}, \alpha[x_1 \mapsto u_1] \cdots [x_n \mapsto u_n] \models \psi.$$

By substitution lemma, which we will prove later

$$\mathfrak{A} \models \psi[x_1 \mapsto t_1] \cdots [x_n \mapsto t_n].$$

$$\mathfrak{A} \models \chi.$$

$$\tau_{\mathfrak{A}} \models_{Gr} \chi.$$

Since $\chi \in \text{Ground}(\Gamma)$ is arbitrarily chosen, we have

$$\forall \chi \in \text{Ground}(\Gamma), \tau_{\mathfrak{A}} \models_{Gr} \chi$$

$$\tau_{\mathfrak{A}} \models_{Gr} \text{Ground}(\Gamma).$$

Therefore $\text{Ground}(\Gamma)$ is propositionally sat. □

Lemma 0.7 (Substitution lemma).

$$\mathfrak{A} \models \psi[x \mapsto t] \iff \mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}}(t)] \models \psi$$

Lemma 0.8.

$$\text{Val}_{\mathfrak{A}, \alpha}(t[x \mapsto t']) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(t)$$

Proof. **Base case:**

1. $t = c \in C$:

$$\text{Val}_{\mathfrak{A}, \alpha}(t[x \mapsto t']) = \text{Val}_{\mathfrak{A}, \alpha}(c) = I(c)$$

$$\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(t) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(c) = I(c)$$

2. $t = y \in V$:

Case 1: $x = y$

$$\implies y[x \mapsto t'] = t'$$

$$\text{Val}_{\mathfrak{A}, \alpha}(y[x \mapsto t']) = \text{Val}_{\mathfrak{A}, \alpha}(t')$$

$$\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(y) = \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')](y) = \text{Val}_{\mathfrak{A}, \alpha}(t')$$

Case 2: $x \neq y$

$$\implies y[x \mapsto t'] = y$$

$$\text{Val}_{\mathfrak{A}, \alpha}(y[x \mapsto t']) = \text{Val}_{\mathfrak{A}, \alpha}(y) = \alpha(y)$$

$$\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(y) = \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')](y) = \alpha(y)$$

Inductive step: $t = f(t_1, \dots)$

$$\begin{aligned} t[x \mapsto t'] &= f(t_1[x \mapsto t'], t_2[x \mapsto t'], \dots) \\ Val_{\mathfrak{A}, \alpha}(t[x \mapsto t']) &= Val_{\mathfrak{A}, \alpha}(f(t_1[x \mapsto t'], t_2[x \mapsto t'], \dots)) = \\ &= I(f)(Val_{\mathfrak{A}, \alpha}(t_1[x \mapsto t']), Val_{\mathfrak{A}, \alpha}(t_2[x \mapsto t']), \dots) \end{aligned}$$

The other side

$$Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t')]}(f(t_1, t_2, \dots)) = I(f)(Val_{\mathfrak{A}, \alpha}(t_1[x \mapsto t']), Val_{\mathfrak{A}, \alpha}(t_2[x \mapsto t']), \dots)$$

□

Proof. We prove the substitution lemma for formulae by structural induction on ψ .

Base case: $\psi \equiv t_1 = t_2$

$$\psi[x \mapsto t] \equiv t_1[x \mapsto t] = t_2[x \mapsto t]$$

Thus

$$\mathfrak{A}, \alpha \models \psi[x \mapsto t] \iff Val_{\mathfrak{A}, \alpha}(t_1[x \mapsto t]) = Val_{\mathfrak{A}, \alpha}(t_2[x \mapsto t]).$$

By the substitution lemma for terms,

$$Val_{\mathfrak{A}, \alpha}(t_1[x \mapsto t]) = Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)]}(t_1)$$

and

$$Val_{\mathfrak{A}, \alpha}(t_2[x \mapsto t]) = Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)]}(t_2).$$

Hence

$$Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)]}(t_1) = Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)]}(t_2),$$

that is,

$$\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] \models t_1 = t_2.$$

Inductive steps

1. $\psi \equiv R(t_1, \dots, t_n)$.

$$\psi[x \mapsto t] \equiv R(t_1[x \mapsto t], \dots, t_n[x \mapsto t]).$$

Then

$$\mathfrak{A}, \alpha \models \psi[x \mapsto t] \iff (Val_{\mathfrak{A}, \alpha}(t_1[x \mapsto t]), \dots, Val_{\mathfrak{A}, \alpha}(t_n[x \mapsto t])) \in I(R).$$

By the substitution lemma for terms, for each i ,

$$\begin{aligned} Val_{\mathfrak{A}, \alpha}(t_i[x \mapsto t]) &= Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)]}(t_i). \\ (Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)]}(t_1), \dots, Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)]}(t_n)) &\in I(R), \\ \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] &\models R(t_1, \dots, t_n). \end{aligned}$$

2. $\psi \equiv \neg\phi$.

$$\begin{aligned} \mathfrak{A}, \alpha \models (\neg\phi)[x \mapsto t] &\iff \mathfrak{A}, \alpha \models \neg(\phi[x \mapsto t]) \\ &\iff \mathfrak{A}, \alpha \not\models \phi[x \mapsto t]. \end{aligned}$$

By IH,

$$\begin{aligned} \mathfrak{A}, \alpha \models \phi[x \mapsto t] &\iff \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] \models \phi. \\ \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] &\not\models \phi, \\ \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] &\models \neg\phi. \end{aligned}$$

3. $\psi \equiv \phi_1 \vee \phi_2$.

$$\begin{aligned} \mathfrak{A}, \alpha \models (\phi_1 \vee \phi_2)[x \mapsto t] &\iff \mathfrak{A}, \alpha \models \phi_1[x \mapsto t] \vee \phi_2[x \mapsto t]. \\ \mathfrak{A}, \alpha \models \phi_1[x \mapsto t] \quad \text{or} \quad \mathfrak{A}, \alpha \models \phi_2[x \mapsto t]. \end{aligned}$$

By IH

$$\begin{aligned} \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] &\models \phi_1 \quad \text{or} \quad \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] \models \phi_2. \\ \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] &\models \phi_1 \vee \phi_2. \end{aligned}$$

□

Question 3

Definition 0.9 (Herbrand Structure). Let $\Sigma = (C, F, R)$ be a signature with $C \neq \emptyset$. The Σ -Herbrand structure is

$$\mathcal{A}_\Sigma = (U_\Sigma, I_\Sigma)$$

defined as follows.

- The universe is

$$U_\Sigma = \text{Terms}_\Sigma$$

(the set of all ground terms).

- Constants are interpreted naturally:

$$I_\Sigma(c) = c.$$

- For every function symbol f of arity r ,

$$I_\Sigma(f)(t_1, \dots, t_r) = f(t_1, \dots, t_r).$$

- Relation symbols may be interpreted arbitrarily.

Lemma 0.10. If $\text{Ground}(\Gamma)$ is propositionally satisfiable then Γ is satisfiable.

Question. Prove Lemma 3.

Proof. Suppose $\text{Ground}(\Gamma)$ is propositionally sat, there is a ground atomic valuation τ

$$\forall \phi \in \text{Ground}(\Gamma), \tau \models_{Gr} \phi$$

Construct a Herbrand structure as per definition. For $I(R)$, we define as such:

$$I_\Sigma^\tau(R) = \{(t_1, t_2, \dots, t_{\text{arity}(R)}) \mid \tau \models_{Gr} R(t_1, t_2, \dots, t_{\text{arity}(R)})\}$$

Now we show that for every ground formula ψ ,

$$\mathfrak{A}_\Sigma^\tau \models \psi \iff \tau \models_{Gr} \psi.$$

Lemma 0.11. For every ground term t ,

$$\text{Val}_{\mathfrak{A}_\Sigma^\tau}(t) = t.$$

Proof. By structural induction on t . If $t \equiv c$ for some constant c , then

$$\text{Val}_{\mathfrak{A}_\Sigma^\tau}(c) = I_\Sigma^\tau(c) = c.$$

If $t \equiv f(t_1, \dots, t_r)$, then by IH

$$\text{Val}_{\mathfrak{A}_\Sigma^\tau}(t_i) = t_i \quad \text{for each } i.$$

$$\text{Val}_{\mathfrak{A}_\Sigma^\tau}(f(t_1, \dots, t_r)) = I_\Sigma^\tau(f)(\text{Val}_{\mathfrak{A}_\Sigma^\tau}(t_1), \dots, \text{Val}_{\mathfrak{A}_\Sigma^\tau}(t_r)) = I_\Sigma^\tau(f)(t_1, \dots, t_r) = f(t_1, \dots, t_r).$$

□

Base cases

1. $\psi \equiv R(t_1, \dots, t_r)$.

By definition,

$$\mathfrak{A}_\Sigma^\tau \models R(t_1, \dots, t_r) \iff (\text{Val}_{\mathfrak{A}_\Sigma^\tau}(t_1), \dots, \text{Val}_{\mathfrak{A}_\Sigma^\tau}(t_r)) \in I_\Sigma^\tau(R).$$

$$(t_1, \dots, t_r) \in I_\Sigma^\tau(R).$$

By definition of $I_\Sigma^\tau(R)$,

$$\tau \models_{Gr} R(t_1, \dots, t_r).$$

Inductive cases

1. $\psi \equiv \neg\phi$.

$$\mathfrak{A}_\Sigma^\tau \models \neg\phi \iff \mathfrak{A}_\Sigma^\tau \not\models \phi.$$

By IH

$$\mathfrak{A}_\Sigma^\tau \not\models \phi \iff \tau \not\models_{Gr} \phi.$$

$$\mathfrak{A}_\Sigma^\tau \models \neg\phi \iff \tau \models_{Gr} \neg\phi.$$

2. $\psi \equiv \phi_1 \vee \phi_2$.

$$\mathfrak{A}_\Sigma^\tau \models \phi_1 \vee \phi_2 \iff \mathfrak{A}_\Sigma^\tau \models \phi_1 \text{ or } \mathfrak{A}_\Sigma^\tau \models \phi_2.$$

By IH

$$\tau \models_{Gr} \phi_1 \text{ or } \tau \models_{Gr} \phi_2.$$

$$\mathfrak{A}_\Sigma^\tau \models \phi_1 \vee \phi_2 \iff \tau \models_{Gr} \phi_1 \vee \phi_2.$$

Now fix a formula

$$\varphi = \forall x_1 \dots \forall x_n \psi \in \Gamma.$$

We show that

$$\mathfrak{A}_\Sigma^\tau \models \varphi$$

fix arbitrary $u_1, \dots, u_n \in U_\Sigma = Terms_\Sigma$ each u_i is a ground term

$$\chi \equiv \psi[x_1 \mapsto u_1] \dots [x_n \mapsto u_n]$$

is a ground instantiation of φ , thus

$$\chi \in Ground(\Gamma).$$

as τ propositionally satisfies $Ground(\Gamma)$

$$\tau \models_{Gr} \chi.$$

$$\mathfrak{A}_\Sigma^\tau \models \chi.$$

$$\mathfrak{A}_\Sigma^\tau \models \psi[x_1 \mapsto u_1] \dots [x_n \mapsto u_n].$$

Since $u_1, \dots, u_n \in U_\Sigma$ are arbitrarily chosen, by the semantics of universal quantifier,

$$\mathfrak{A}_\Sigma^\tau \models \forall x_1 \dots \forall x_n \psi.$$

$$\mathfrak{A}_\Sigma^\tau \models \varphi.$$

Since $\varphi \in \Gamma$ is also arbitrarily chosen, we have

$$\mathfrak{A}_\Sigma^\tau \models \Gamma.$$

□

Question 4

Theorem 0.12 (Downward Löwenheim–Skolem Theorem). *Let Σ be a countable signature and let Γ be a set of universally quantified FOL sentences without equality.*

If Γ is satisfiable, then there exists a countable structure $\mathcal{A}$ such that

$$\mathcal{A} \models \Gamma.$$

Question. Prove Theorem 2.

Proof. Suppose Γ is satisfiable, $Ground(\Gamma)$ is propositionally sat, there exists a Herbrand structure

$$\mathfrak{A}_\Sigma^\tau = (U_\Sigma, I_\Sigma^\tau)$$

such that

$$\mathfrak{A}_\Sigma^\tau \models \Gamma.$$

Easy to show that $\mathfrak{A}_\Sigma^\tau$ is countable

$$U_\Sigma = Terms_\Sigma,$$

the set of all ground terms over the signature Σ . Since Σ is countable, and every ground term is a finite string over the symbols of Σ , the set $Terms_\Sigma$ is countable.

Therefore $\mathfrak{A}_\Sigma^\tau$ is a countable structure satisfying Γ . □

Question 5

Theorem 0.13. Γ is satisfiable if and only if $\text{Ground}(\Gamma)$ is satisfiable.

Two sets of sentences Γ_1, Γ_2 are said to be *equivalent* if for every structure $\mathcal{A}$,

$$\mathcal{A} \models \Gamma_1 \iff \mathcal{A} \models \Gamma_2.$$

Question.

Describe:

- a signature Σ ,
- a set Γ of universally quantified FOL sentences without equality over Σ ,
- a Σ -structure $\mathcal{A}$,

such that

$$\mathcal{A} \models \text{Ground}(\Gamma) \quad \text{but} \quad \mathcal{A} \not\models \Gamma.$$

Proof. Consider signature

$$\Sigma = (\{0\}, \emptyset, \{R^1\})$$

structure

$$\mathfrak{A} = (\{0, 1\}, I(0) = 0, I(R) = \{0\})$$

singleton set

$$\Gamma = \{\forall x R(x)\}$$

$$\text{Ground}(\Gamma) = \{R(0)\}$$

$$\mathfrak{A} \models \text{Ground}(\Gamma)$$

but

$$\mathfrak{A} \not\models \Gamma$$

as $R(1) \notin I(R)$. Basically grounding loses information about elements in the universe that no ground terms represent $\square$

Question 6

Let

$$\Sigma = (\{c, d, e\}, \{f\}, \{R\})$$

where f is unary and R is unary.

Define the formulas

$$\psi_1 \equiv \neg(c = c)$$

$$\psi_2 \equiv (c = d) \wedge \neg(d = c)$$

$$\psi_3 \equiv (c = d) \wedge (d = e) \wedge \neg(c = e)$$

$$\psi_4 \equiv (c = d) \wedge \neg(f(c) = f(d))$$

$$\psi_5 \equiv (c = d) \wedge R(c) \wedge \neg R(d)$$

Let

$$\Gamma_i = \{\psi_i\}$$

for $i \in \{1, 2, 3, 4, 5\}$.

Question.

Show that for each i ,

$$\text{Ground}(\Gamma_i)$$

is propositionally satisfiable but

$$\Gamma_i$$

is not satisfiable.

1. $\psi_1 \equiv \neg(c = c)$

- Propositionally satisfiable: let $\tau(c = c) = \text{false}$.
- Not satisfiable in FOL: equality is reflexive, $c = c$

2. $\psi_2 \equiv (c = d) \wedge \neg(d = c)$

- Propositionally satisfiable: let $\tau(c = d) = \text{true}$ and $\tau(d = c) = \text{false}$.
- Not satisfiable in FOL: equality is symmetric, $c = d \Rightarrow d = c$.

3. $\psi_3 \equiv (c = d) \wedge (d = e) \wedge \neg(c = e)$

- Propositionally satisfiable: let $\tau(c = d) = \tau(d = e) = \text{true}$ and $\tau(c = e) = \text{false}$.
- Not satisfiable in FOL: equality is transitive, $c = d \wedge d = e \Rightarrow c = e$.

4. $\psi_4 \equiv (c = d) \wedge \neg(f(c) = f(d))$

- Propositionally satisfiable: let $\tau(c = d) = \text{true}$ and $\tau(f(c) = f(d)) = \text{false}$.
- Not satisfiable in FOL: functions preserve equality, $c = d \Rightarrow f(c) = f(d)$.

5. $\psi_5 \equiv (c = d) \wedge R(c) \wedge \neg R(d)$

- Propositionally satisfiable: let $\tau(c = d) = \text{true}$, $\tau(R(c)) = \text{true}$, $\tau(R(d)) = \text{false}$.
- Not satisfiable in FOL: relations preserve equality, $c = d \Rightarrow (R(c) \leftrightarrow R(d))$.

Question 7

Let

$$\Gamma^* = eq(\Gamma) \cup E$$

where E is the set of equality axioms.

Lemma 0.14. *If Γ is satisfiable then Γ^* is satisfiable.*

Question. Prove Lemma 4.

Proof. Intuition: $\Gamma \mapsto \Gamma^*$ just constraints the interpretation of relation EQ, therefore under any structure, if a formula $\phi \in \Gamma$ is true, corresponding $eq(\phi) \in \Gamma^*$ is true

Suppose Γ is sat, then there exists a structure $\mathfrak{A} = (U, I)$ such that $\mathfrak{A} \models \Gamma$.

Define a new structure $\mathfrak{A}' = (U, I')$ based on $\mathfrak{A}$ as such

- For all symbols in Σ , interpretation remains the same
- Interpret EQ as

$$I'(EQ) = \{(u, u) \mid u \in U\}.$$

Then $\mathfrak{A}'$ satisfies all equality constraints E , as identity is an equivalence relation and is preserved by functions and relations. also, since $eq(\Gamma)$ is obtained from Γ by replacing equality with EQ , and EQ is interpreted as identity, trivially

$$\begin{aligned}\mathfrak{A}' &\models eq(\Gamma). \\ \mathfrak{A}' &\models eq(\Gamma) \cup E = \Gamma^*.\end{aligned}$$

□

Question 8

Definition 0.15 (Equivalence Relation). *A binary relation $R \subseteq U \times U$ is an equivalence relation if it is*

- *reflexive*
- *symmetric*
- *transitive*

For $u \in U$ the equivalence class is

$$[u]_R = \{v \mid (u, v) \in R\}.$$

The quotient set is

$$U/R = \{[u]_R \mid u \in U\}.$$

Definition 0.16 (Congruence Relation). *Let U be a set and $E \subseteq U \times U$. E is a congruence relation with respect to functions F and relations R if*

1. *E is an equivalence relation.*
2. *For every function f of arity r ,*

$$E(u_1, v_1), \dots, E(u_r, v_r) \Rightarrow E(f(u_1, \dots, u_r), f(v_1, \dots, v_r)).$$

3. *For every relation R of arity r ,*

$$E(u_1, v_1), \dots, E(u_r, v_r)$$

implies the tuples either both belong to R or both do not.

Question.

Let $\mathcal{A} = (U, I)$ be a structure for the signature

$$\Sigma^* = (C, F, R \cup \{EQ\})$$

such that

$$\mathcal{A} \models E.$$

Show that $I(EQ)$ is a congruence relation with respect to every function symbol $f \in F$ and relation symbol $R \in R$.

Proof. since $\mathcal{A} \models E_1, E_2, E_3$, the relation $I(EQ)$ is reflexive, symmetric, and transitive, therefore $I(EQ)$ is an equivalence relation.

let $f \in F$ be of arity r , and let $u_1, \dots, u_r, v_1, \dots, v_r \in U$ satisfy

$$(u_i, v_i) \in I(EQ) \quad (1 \leq i \leq r).$$

Since $\mathcal{A} \models E_4$, it follows that

$$(f(u_1, \dots, u_r), f(v_1, \dots, v_r)) \in I(EQ).$$

Thus $I(EQ)$ is preserved by every function symbol.

Similarly for relation symbols we have

$$R(u_1, \dots, u_r) \iff R(v_1, \dots, v_r).$$

Hence $(u_1, \dots, u_r) \in I(R)$ iff $(v_1, \dots, v_r) \in I(R)$, so $I(EQ)$ is consistent with every relation symbol.

Therefore $I(EQ)$ is a congruence relation with respect to every function symbol in F and every relation symbol in $\mathcal{R}$. $\square$

Question 9

Lemma 0.17. *If Γ^* is satisfiable then Γ is satisfiable.*

Question. Prove Lemma 5. Heavy duty at work this 2 weeks, unable to finish, will attempt alongside with hw5

Question 10

Theorem 0.18. *Assume that the signature Σ contains at least one constant.*

Let Γ be a set of universally quantified FOL sentences in prenex normal form.

Then

$$\Gamma \text{ is satisfiable} \iff \text{Ground}(\Gamma^*) \text{ is propositionally satisfiable.}$$

Question. Prove Theorem 5.